

1) Caesar cipher

a) Encrypt "Defend The wall" with key = 3
shift each letter by +3

Plain text

Defend The wall

Cipher text

G I J H Q L N R P T W = 000



b) Decrypt "KHOOZ ZRLOG" with key = 3

Shift each letter by -3

Cipher text

KHOOZ ZRLOG

Plain text

Hello world

c) Secret → VHFUHR

Caesar cipher : S - 6 (73)

Therefore key = -3

Decrypt

W K C C L L O R W H U

Plain text

This is A TEST

2) monoalphabetic substitution cipher:

a) Encrypt "FIRE"

A B C D E F G H I J K L M N O P Q R S
T U V W X Y Z

A B C D E F G H I J K L M N O P Q R S
T U V W X Y Z

Encryption

cipher text

F → Y

Y O K I

I → O

R → K

E → T

b) Decrypt:

cipher text:

Y O K I

~~Q E B~~

N R F Z H

Y O K I

plain text:

~~THE~~ Q L T C R BROWN FOX

(Approximate English sentence obtained using frequency)

3) Vigenere cipher

key used:

LEMON

a) Encrypt

Plain text

ATTACK AT PALACE

Repeat key

LEMON LEMON LEMON

cipher text

L X F O P L E R N H R M T I

b) Decrypt

ciphertext

CXFOPLERWHR

keyword

LEMON

plaintext

ACKACK ATOWWW

c) Decrypt

ciphertext

QWXPKNAGTU

keyword:

ICEY

plaintext:

WOTATACWAPK

4) Question 4 - Hill cipher

use matrix $K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$

a) Encrypt "HI"

H = 7

I = 8

vector

$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$

multiply

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

modulo 26

$$45 \text{ mod } 26 = 19 = 7$$

$$54 \text{ mod } 26 = 2 = 0$$

~~cipher~~

ciphertext

TC

b) Verify P0 → Q1

P = 15

Q = 14

multiply

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 102 \end{bmatrix}$$

modulus 26
 $87 \bmod 26 = 9 = 1$
 $100 \bmod 26 = 22 = 10$

Result
 54

5) S-playfair cipher.

Key word
 ZEBRAS

a) Encrypt.

plaintext

MEETME AFTERNOON

ciphertext

NFKLVFBGSKPODLVR

Result

BOHR O

b) Decrypt:

ciphertext

GAMTFFCCHETREL

plaintext

MEETME AFTERNOON

3)

6) one Time Pad

a) Encrypt

plaintext

hello

key

xmckl

convert to numbers

H = 7	X = 23
E = 4	M = 12
L = 11	C = 2
O = 14	K = 10
	L = 11

Full mod 26

LEMON LEMONLEMON

$$7+23=30 \rightarrow 4=C$$

$$4+12=16=A$$

$$11+2=13=M$$

$$11+10=21=U$$

$$14+11=25=Z$$

Cyphs :-

CA MR

b) why OTP is unbreakable

key length equal messages length

key is completely random

key is used only one

key random level

7) Question - columnar Transposition

a) Encrypt

Plaintext

WE ARE U IS POWER

Key used

ZEBRA

Arrange:

Z E B R A

W E A R U

U I S C O

V E R E D

column order

A B E R Z

5 3 2 4 1

Cyphs is

E O O A S R G I C E K V D

b) Decrypt

Cyphs is

E U L N E A C O T K E S C A Q R O P E C O X W W

Key used:

OTR I A L

Received plain list

WEA R 6035 C0K0 VEG FLEE A TONCE

Encrypt.

Plain list

INFORMATION SECURITY

Key material:

HACK

length = 19

column = 4

Row

$19 \times 4 = 76$ determine 3

Row needed = 5

Cipher list

O T E I I F O T R R M A Y V V S C O T

8) Frequency Analysis.

Cipher list

V Z Q S O U O H X M O P V G P E V S G W Z O P F P G S X V Z V U G

Frequency

Z → 8
O → 7
V → 3
P → 3
S → 4
E → 4

possible mapping

Z → C
O → T
P → A
V → O
S → I

partial

DE...TO

9 known plaintext

cipher text

BSTW QTW ymj w N F xNRVQSW

known plaintext

this is the

mapping

B → T

T → H

W → I

Q → S

V → G

M → T

N → I

X → S

10 vigenere cipher

cipher text

ZIC VTWQ. NGR ZKVT KAVZHC QY4CMB

plaintext

DECEPTIVE

decrypted plaintext

WE ARE ISLAND CREEPS SAVE YOUR SEED

on